

PHY206 Final Practice Question – Spring 2026

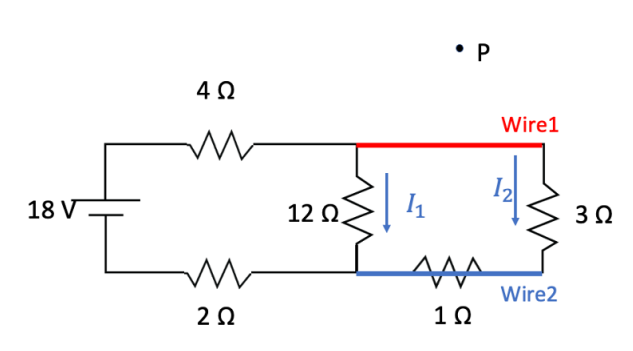
Answer the questions in the spaces provided. For any calculations, you need to clearly show your work and steps. If you run out of room for an answer, continue on the back of the page (and clearly mark the continued location).

Name: _____

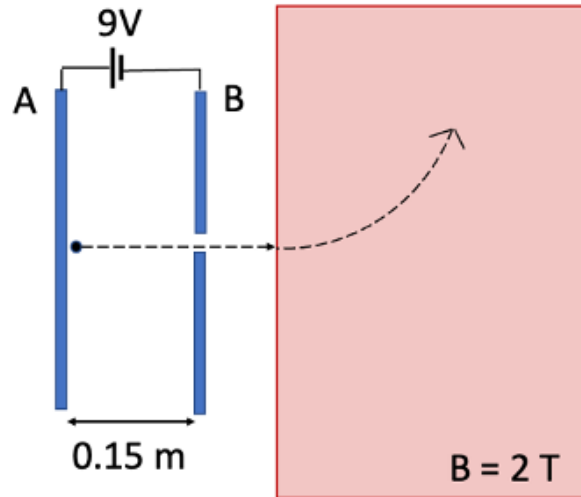
1. A storm cloud 200 m off the ground sits over a fish pond that is initially at 20°C . The pond is defined to be at voltage zero and the cloud is at 250 MV. A lightning strike carrying 50 C of charge leaves the cloud and hits the pond.
 - (a) What is the electric potential energy carried by the 50 C of charge?
 - (b) Assuming that all the energy of the transferred charge is absorbed by the water, how many kg of water can be raised to 100°C and then boiled away?

2. A circuit is set up as the figure below.

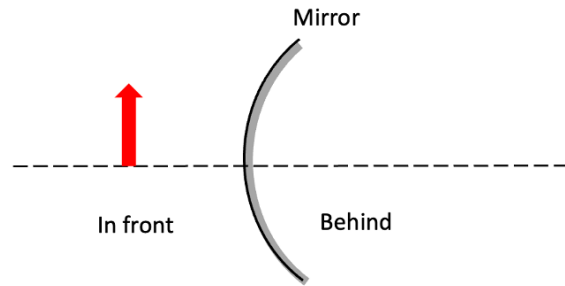
- (a) Find current I_1 and I_2 .
- (b) If the $3\ \Omega$ resistor is a heater and is heating a cup of 30 g of water and 10 g of ice that is originally at 0°C . What would be the final temperature of the water after 30 minutes?
- (c) Draw two arrows at location P to show the direction of magnetic field due to Wire1 and Wire2, respectively.



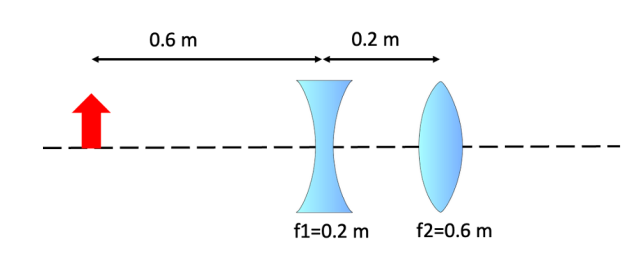
3. Two very large conductor plates are connected to a 9-V battery, as shown in the figure. A proton is released from plate A, passes through a tiny slit in plate B and enters an area with a uniform magnetic field of 2 T (the red area shown in the plot). When entering the magnetic field area, the proton starts to curve upward. The dashed line in the figure shows the proton's trajectory. You can assume the gravitational force on the proton is very small and can be ignored.
- (a) What are the surface charge densities carried by plate A and plate B, respectively? (Remember to note whether the charge is positive or negative.)
 - (b) What are the direction and magnitude of the electric force experienced by the proton when it is in between plate A and B?
 - (c) What are the direction and magnitude of the magnetic force when the proton just enters the red area? The magnetic field in this area is pointing 90° from proton travel direction.
 - (d) What is the direction of the magnetic field in the red area?



4. (a) An object that is 3 cm tall is placed at 10 cm in front of a convex mirror, as shown in figure below. The mirror has a focal length of 20 cm. What are the location and the height of the image? Is the image real or virtual? Is the image upright or inverted? For the location of the image, make sure you state the distance of the image to the mirror, and whether it is in front or behind the mirror.



- (b) Two thin lenses of focal lengths $f_1 = 0.2 \text{ m}$ and $f_2 = 0.6 \text{ m}$ are separated by 0.2 m . An object is placed 0.6 m to the left of lens 1. Find the position of the final image of the system.



Here are the generic equations. If you use any equations other than these, please clearly show the steps of how you get there.

$$\Delta L = \alpha L_i \Delta T \quad PV = nRT$$

$$\Delta E_{\text{int}} = Q - W \quad \langle \text{K.E.} \rangle = \frac{3}{2} kT \quad W = \int_{V_i}^{V_f} P dV \quad W = P(V_f - V_i) \quad W = nRT \ln\left(\frac{V_f}{V_i}\right)$$

$$Q = mc\Delta T \quad Q = mL \quad Q = nC_p\Delta T \quad Q = nC_v\Delta T \quad e = \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h}$$

$$E = \frac{kQ}{r^2} \quad F = \frac{kQq}{r^2} \quad E = \frac{\sigma}{2\epsilon_0} \quad E = \frac{\sigma}{\epsilon_0} \quad F = qE$$

$$\Delta V = - \int \vec{E} \cdot d\vec{s} = \frac{\Delta PE_E}{q} \quad \text{K.E} = \frac{1}{2}mv^2 \quad \Phi_E = \int_{\text{surface}} \vec{E} \cdot d\vec{A}$$

$$\text{Gauss's law: } \Phi_E = \oint_{\text{surface}} \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

$$V = IR \quad P = IV$$

$$\text{In series: } R_{\text{eq}} = R_1 + R_2 + \dots$$

$$\text{In parallel: } \frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$\text{RC circuit: } Q(t) = C\mathcal{E}(1 - e^{-t/RC}) \quad I(t) = \frac{\mathcal{E}}{R}e^{-t/RC} \quad Q(t) = Q_i e^{-t/RC} \quad I(t) = -\frac{Q_i}{RC}e^{-t/RC}$$

$$F = ma \quad \vec{F}_B = q\vec{v} \times \vec{B} \quad F_B = qvB \sin\theta \quad \vec{F}_B = I\vec{L} \times \vec{B} \quad F_B = ILB \sin\theta$$

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \quad \Phi_B = \int \vec{B} \cdot d\vec{A} \quad |\mathcal{E}| = B\ell v \quad V = IR \quad P = IV$$

$$B = \frac{\mu_0 NI}{L} \quad B = \frac{\mu_0 I}{2a} \quad B = \frac{\mu_0 I}{2\pi a} \quad \mathcal{E} = -L \frac{dI}{dt} \quad L = \frac{N\Phi_B}{I}$$

Numbers you might need:

- Boiling point of water: 100°C; melting point of ice: 0°C.
- Specific heat of ice: 2050 J/(kg · C); specific heat of water: 4186 J/(kg · C)
- Latent heat of fusion for water: 3.33×10^5 J/kg
- Latent heat of vaporization for water: 2.26×10^6 J/kg
- 1 kg = 1000 g; $R = 8.31$ J/(mol · K); Avogadro's number = 6.022×10^{23}
- Boltzmann constant $k = 1.38 \times 10^{-23}$ J/K
- Charge of a proton: 1.6×10^{-19} C; Charge of an electron: -1.6×10^{-19} C
- Mass of a proton: 1.67×10^{-27} kg; Mass of an electron: 9.11×10^{-31} kg
- Coulomb's constant: $k = 8.99 \times 10^9$ [N · m²/C²]
- $\epsilon_0 = 8.85 \times 10^{-12}$ [C/(N · m²)]; $\mu_0 = 4\pi \times 10^{-7}$ [T · m/A]
- Speed of light in vacuum $c = 3 \times 10^8$ m/s

Signs for mirrors:

	Positive value (+)	Negative value (-)
d_o	Real/front	Virtual/back
d_i	Real/front	Virtual/back
f, R	Converging mirror	Diverging mirror

Signs Conventions for thin lenses

- d_o is positive if object is in front of lens (real object).
- d_o is negative if object is in back of lens (virtual object).
- d_i is positive if image is in back of lens (real image).
- d_i is negative if image is in front of lens (virtual image).
- f is positive if the lens is converging lens.
- f is negative if the lens is diverging lens.